Zahra Khatti

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About Me

Ph.D. student in Industrial and Systems Engineering at Lehigh University specializing in large-scale stochastic optimization. Research focuses on algorithm design for adaptive sampling and fairness-aware machine learning, supported by a strong foundation in mathematical analysis, optimization under constraints, and neural network implementation. Proficient in Python and PyTorch, with research interests spanning optimization algorithms, mathematical modeling, applied machine learning, and forecasting.

Skills

- **Mathematical:** Optimization, Operations Research, Machine Learning, Probability, Statistics
- **Computational:** Python, PyTorch, Matlab, Gurobi, AMPL, R, Git, Arena, GAMS, Numpy, Pandas

Experience

Research Assistant — **Aug. 2023 – Present**
Lehigh University, Department of Industrial and Systems Engineering — *Bethlehem, PA*

– Conducting research on **adaptive sampling methods** for stochastic optimization, focusing on dynamically adjusting sample sizes across iterations to improve efficiency and convergence guarantees.
– Developing theoretical proofs on algorithm convergence under various assumptions and analyzing bounds.
– Designed a new fairness measure and applied it to large-scale machine learning problems with nonconvex constraint-based formulations.
– Implemented neural networks in **PyTorch** and applied **SQP-based optimization** for constrained machine learning.

Marketing Strategist — **Jan. 2018 – Jul. 2018**
K. N. Toosi Innovation Center — *Tehran, Iran*

– Led marketing strategies for a sports-focused startup, increasing user engagement and brand awareness.
– Analyzed market trends and customer feedback to guide strategic decision-making.

Education

Lehigh University — **Aug. 2023 – Present**
PhD in Industrial and Systems Engineering — *Bethlehem, PA*

– Advisors: Prof. Frank E. Curtis, Prof. Daniel P. Robinson
  - Research Focus: Large-Scale Optimization Algorithms for Nonlinear Problems with Applications in Machine Learning and Data Science.

Sharif University of Technology — **2019 – 2022**
Master of Science in Industrial Engineering, Economic and Social Systems — *Tehran, Iran*

– Advisor: Prof. Farhad Kianfar
  - Dissertation: Customer Segmentation and Migration Pattern Modeling Based on Loyalty.

K.N. Toosi University of Technology — **2015 – 2019**
Bachelor of Science in Industrial Engineering — *Tehran, Iran*

Publications

1. **Zahra Khatti, Daniel P. Robinson, and Frank E. Curtis.** Fair supervised learning through constraints on smooth nonconvex unfairness-measure surrogates. *arXiv preprint arXiv:2505.15788*, 2025.

Leadership and Service

MOPTA Conference Co-chair — **2026 (ongoing)**
Lehigh University — *Bethlehem, PA*

– Co-organizing the 2026 MOPTA conference; coordinating speakers, sessions, and logistics.

GradOpt and ColOpt Workshops Co-chair — **2026 (ongoing)**
Lehigh University — *Bethlehem, PA*

– Organizing workshops, including inviting junior faculty for ColOpt and master's/undergraduate students for GradOpt.

Graduation Ceremony Organizer — **2019**
K.N. Toosi University of Technology — *Tehran, Iran*

Executive Board Member — **2016–2017**
Scientific Association, K.N. Toosi University — *Tehran, Iran*

Selected Courses

Machine Learning · Neural Networks, Convolutional Neural Networks, Optimization Algorithms (SGD, Adam), Convergence Theorems
Statistical Machine Learning · Supervised Learning, Linear & Logistic Regression, Decision Trees, SVM, Ensemble Methods (Random Forest, AdaBoost, XGBoost)
Continuous Optimization · Gradient-based Methods, Constrained Optimization, Nonlinear Programming, Lagrangian Methods
Project Management · Planning, Scheduling, Risk Management, Resource Allocation
Decision Management · Bayesian Decision Analysis, Multi-criteria Decision Analysis
Integer Programming (IP) · Branch and Bound, Column Generation, Heuristics

Presentations

Presentation — **Dec. 2025 (upcoming)**
NeurIPS 2025 Workshop on Constraint Optimization in Machine Learning (COML) — *San Diego, CA*
- Accepted to present research poster on *"Fair Supervised Learning Through Constraints on Smooth Nonconvex Unfairness-Measure Surrogates"*.

Honors and Awards

- Full Scholarship from Sharif University (M.Sc.)
- Selected as an "Exceptionally Talented" Student, Department of Industrial Engineering, Sharif University
- Ranked 3rd among 42 students in B.Sc., Department of Industrial Engineering, K.N. Toosi University
- Member of the Scientific Association, Winner of the 10th "Movement" Festival
- Full Scholarship from K.N. Toosi University (B.Sc.)

Selected Projects

Review of Uncertainty Quantification for Deep Learning Models — **2023**
Lehigh University — *Bethlehem, PA*
- Reviewed and analyzed a research paper on uncertainty quantification techniques for deep learning regression models under limited data conditions.

Business Process Simulation Models from Event Logs — **2020**
Sharif University of Technology — *Tehran, Iran*
- Developed a model for automated discovery of business process simulation models based on event log data.

Multi-Stage Financial Planning Problem — **2020**
Sharif University of Technology — *Tehran, Iran*
- Applied scenario reduction and nested decomposition methods to optimize a multi-stage financial planning problem.